Proof of Parseval's Theorem

Goal:
We want to prove that the total energy of a discrete signal is the same in its spatial/time domain and its frequency domain. Given the unitary Discrete Fourier Transform (DFT) definitions:

- Forward DFT: $F(\omega) = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} f(x) e^{-i\omega x}$
- Inverse DFT: $f(x) = \frac{1}{\sqrt{N}} \sum_{\omega} F(\omega) e^{i\omega x}$

We must prove:
$\sum_{x=0}^{N-1} |f(x)|^2 = \sum_{\omega} |F(\omega)|^2$

Step-by-Step Proof:

Step 1: Start with the Left-Hand Side (LHS)

We begin with the sum of the squared magnitudes of the signal in the spatial domain.
LHS = $\sum_{x=0}^{N-1} |f(x)|^2$

Step 2: Express Magnitude Squared as a Product

A fundamental property of complex numbers is that the squared magnitude of a number z is equal to the product of the number and its complex conjugate, i.e., $|z|^2 = z \cdot z^*$. We apply this to $f(x)$.

LHS = $\sum_{x=0}^{N-1} f(x) \cdot f^*(x)$

Step 3: Substitute the Inverse DFT Formula

Now, we replace $f(x)$ with its inverse DFT definition. To avoid confusion with summation variables later, we will also write out the expression for the complex conjugate $f^*(x)$.

- The inverse DFT for $f(x)$ is:
  $f(x) = \frac{1}{\sqrt{N}} \sum_{\omega} F(\omega) e^{i\omega x}$
- To find $f^*(x)$, we take the complex conjugate of the entire expression. We use a different variable for the frequency summation, $\omega^{'}$, to keep the sums distinct.
- By using two **distinct variables** (ω and $\omega^{'}$), we make it mathematically clear that the index of the first sum and the index of the second sum are **independent** and must be treated separately.
  $f^*(x) = (\frac{1}{\sqrt{N}} \sum_{\omega^{'}} F(\omega^{'}) e^{i\omega^{'} x})^* = \frac{1}{\sqrt{N}} \sum_{\omega^{'}} F^*(\omega^{'}) e^{-i\omega^{'} x}$

Substituting both into our equation:

LHS = $\sum_{x=0}^{N-1} \left(\frac{1}{\sqrt{N}} \sum_{\omega} F(\omega)e^{i\omega x}\right)\left(\frac{1}{\sqrt{N}} \sum_{\omega'} F^*(\omega')e^{-i\omega' x}\right)$

Step 4: Rearrange the Terms and Summations

We can combine the constants and rearrange the order of the summations. Since the sums are finite, we can swap their order. We group all terms that depend on x together in the innermost sum.

LHS = $\frac{1}{N} \sum_{\omega} \sum_{\omega'} F(\omega)F^*(\omega') \sum_{x=0}^{N-1} e^{i\omega x}e^{-i\omega' x}$

LHS = $\frac{1}{N} \sum_{\omega} \sum_{\omega'} F(\omega)F^*(\omega') \left(\sum_{x=0}^{N-1} e^{i(\omega-\omega')x}\right)$

Step 5: Evaluate the Innermost Sum (Orthogonality Property)

The key to the proof lies in evaluating the sum over x. This sum represents the orthogonality property of the complex exponential basis functions of the DFT. Let's analyze it in two cases.

- **Case 1:** $\omega = \omega'$
  If the frequencies are the same, the exponent becomes zero:
  $e^{i(\omega-\omega)x} = e^0 = 1$
  The sum becomes: $\sum_{x=0}^{N-1} 1 = N$
- **Case 2:** $\omega \neq \omega'$
  The sum is a finite geometric series $\sum_{j=0}^{N-1} r^j$ with $r = e^{i(\omega-\omega')}$. The sum of such a series is $\frac{1-r^N}{1-r}$.
  The sum is $\frac{1-(e^{i(\omega-\omega')})^N}{1-e^{i(\omega-\omega')}} = \frac{1-e^{i(\omega N-\omega' N)}}{1-e^{i(\omega-\omega')}}$.
  Since $\omega = \frac{2\pi k}{N}$ for an integer k, then $\omega N = 2\pi k$. Thus, $e^{i\omega N} = e^{i2\pi k} = 1$. Similarly, $e^{i\omega' N} = 1$.
  The numerator becomes $1 - (1 \cdot 1) = 0$. Since $\omega \neq \omega'$, the denominator is not zero.
  Therefore, the sum is 0.

  What is `j` and why was the variable `x` replaced?

The original sum in the proof is:
$\sum_{x=0}^{N-1} e^{i(\omega-\omega')x}$

This can be rewritten by grouping the base of the exponent:
$\sum_{x=0}^{N-1} (e^{i(\omega-\omega')})^x$

The text then re-writes this in the general form $\sum_{j=0}^{N-1} r^j$. This is a common technique in mathematical proofs to make it clear that a specific expression matches a general, well-known formula.

- **What is `j`?**
  `j` is a **dummy index variable** for the summation. Its role is identical to the role

that `x` was playing. It is simply a counter that takes on the integer values from 0 up to N-1.

- **Why was `x` replaced with `j`?**
  The variable was changed from `x` to `j` purely for **clarity and convention**. The standard formula for a finite geometric series is almost always written with an index like `j`, `k`, or `n`.

  By making the substitutions:

  - $r = e^{i(\omega - \omega')}$
  - $j = x$

  ...the author is explicitly showing that our specific, complicated-looking sum is nothing more than a standard geometric series. This makes it immediately obvious which formula to apply next.

2. How did the sum of the series come from?

The formula for the sum of a finite geometric series is a classic mathematical result. Here is a short derivation of how it is found.

Let the sum be called S. The series is:
$S = \sum_{j=0}^{N-1} r^j = 1 + r + r^2 + ... + r^{N-1}$

Now, follow these steps:

Multiply the entire series by r:
$rS = r(1 + r + r^2 + ... + r^{N-1})$
$rS = r + r^2 + r^3 + ... + r^N$

Subtract the second equation from the first one ($S - rS$):
$S - rS = (1 + r + r^2 + ... + r^{N-1}) - (r + r^2 + r^3 + ... + r^N)$

Notice that almost all the terms cancel out:
The `$r$` in the first part cancels with the `$r$` in the second part. The `$r^2$` cancels with `$r^2$`, and so on, until `$r^{N-1}$` is cancelled. The only terms that do **not** cancel are the first term of S (which is `1`) and the last term of rS (which is `$r^N$`).

This leaves us with:
$S - rS = 1 - r^N$

Solve for S:
Factor out S on the left side:
$S(1 - r) = 1 - r^N$

Divide by `(1 - r)` to isolate S:
$S = \frac{1-r^N}{1-r}$

The foundation for this is Euler's formula, which connects the complex exponential function to trigonometric functions:

$e^{i\theta} = \cos(\theta) + i\sin(\theta)$

This formula tells us that the number $e^{i\theta}$ can be visualized as a point on a circle with a radius of 1 (the "unit circle") in the complex plane. The angle θ determines where on that circle the point is located.

Applying Euler's Formula to the Problem

Identify the Angle (θ)
In our expression, $e^{i2\pi k}$, the angle is $\theta = 2\pi k$.

Substitute into Euler's Formula
$e^{i2\pi k} = \cos(2\pi k) + i\sin(2\pi k)$

Evaluate the Cosine and Sine Terms
We know that `k` **is an integer** (k = 0, 1, 2, 3, ...). Let's see what happens to the cosine and sine for different integer values of `k`:

- **If k = 0:** $\cos(0) = 1$ and $\sin(0) = 0$
- **If k = 1:** $\cos(2\pi) = 1$ and $\sin(2\pi) = 0$
- **If k = 2:** $\cos(4\pi) = 1$ and $\sin(4\pi) = 0$
- **If k = -1:** $\cos(-2\pi) = 1$ and $\sin(-2\pi) = 0$

An angle of 2π represents one full rotation around a circle. Therefore, an angle of $2\pi k$ represents `k` **full rotations**. No matter how many full rotations you make, you always end up back at the starting point.

On the unit circle, the starting point (an angle of 0) corresponds to the coordinates (1, 0).

- The cosine value is the x-coordinate, which is always **1**.
- The sine value is the y-coordinate, which is always **0**.

Therefore, for any integer `k`:

- $\cos(2\pi k) = 1$
- $\sin(2\pi k) = 0$

Calculate the Final Result
Now we plug these values back into our equation:

$e^{i2\pi k} = 1 + i(0)$
$e^{i2\pi k} = 1$

Summary

So, the inner sum is N if $\omega = \omega'$ and 0 otherwise. This can be written concisely using the Kronecker delta function, $\delta_{\omega,\omega'}$:
$\sum_{x=0}^{N-1} e^{i(\omega-\omega')x} = N \cdot \delta_{\omega,\omega'}$

Step 6: Simplify the Main Expression

We substitute this result back into our main equation from Step 4.

LHS = $\frac{1}{N} \sum_{\omega} \sum_{\omega'} F(\omega)F^*(\omega')(N \cdot \delta_{\omega,\omega'})$

The N in the numerator and denominator cancel out:

LHS = $\sum_{\omega} \sum_{\omega'} F(\omega)F^*(\omega')\delta_{\omega,\omega'}$

Now we evaluate the sum over ω'. The Kronecker delta $\delta_{\omega,\omega'}$ ensures that the only term that is not zero is the one where $\omega' = \omega$. This effectively collapses the inner sum, and we replace every instance of ω' with ω.

LHS = $\sum_{\omega} F(\omega)F^*(\omega)$

Step 7: Final Step

Finally, we use the same property from Step 2, $z \cdot z^* = |z|^2$, this time for the frequency domain function $F(\omega)$.

LHS = $\sum_{\omega} |F(\omega)|^2$

This is the Right-Hand Side (RHS) of the theorem. We have successfully shown that:

$\sum_{x=0}^{N-1} |f(x)|^2 = \sum_{\omega} |F(\omega)|^2$

Q.E.D.